Machine learning

8305041-2

Course overview

Machine learning through concepts, worked examples, implementation, and careful evaluation.

HOW WE LEARN

- Lectures and class notes
- Short worked exercises
- Self-training and implementation

OPTIONAL READING

- Bishop — Pattern Recognition and Machine Learning
- Goodfellow et al. — Deep Learning
- Witten et al. — Data Mining

ASSESSMENT

- 10–15 quizzes and exams
- 5–10 self-training or project tasks

Course habit: understand the idea → implement it → evaluate it honestly.

Basic Concepts

Basic Concepts

How observed examples become predictions for new cases

Everyday learning: two paths

Feedback

An adult points to an animal:
‘That is a cat.’

Answer supplied

Names, grades, coaching

Patterns

You notice which songs are often played together.

No answer supplied

Groups, habits, similarities

Machine learning follows the same two patterns

Supervised learning

Past apartments include features
and a known outcome.

Learn a rule to predict a target
Rent amount / rented quickly?

Unsupervised learning

Apartment descriptions have features
but no recorded outcome.

Find structure in the data

Similar-apartment groups

Supervised learning: two kinds of target

Regression

How much will the apartment's monthly rent be?

A numerical target

Example: 1600

Classification

Will the apartment rent within one week?

A categorical target

Within one week / later

Supervised learning uses examples with known targets.

Inputs, targets, and observations

Feature (Recorded property)	**Value**
Floor area	70 m²
Bedrooms	2
Neighborhood	Amir Abad
Target to predict: **Monthly Rent**	**1500**

One instance

An apartment

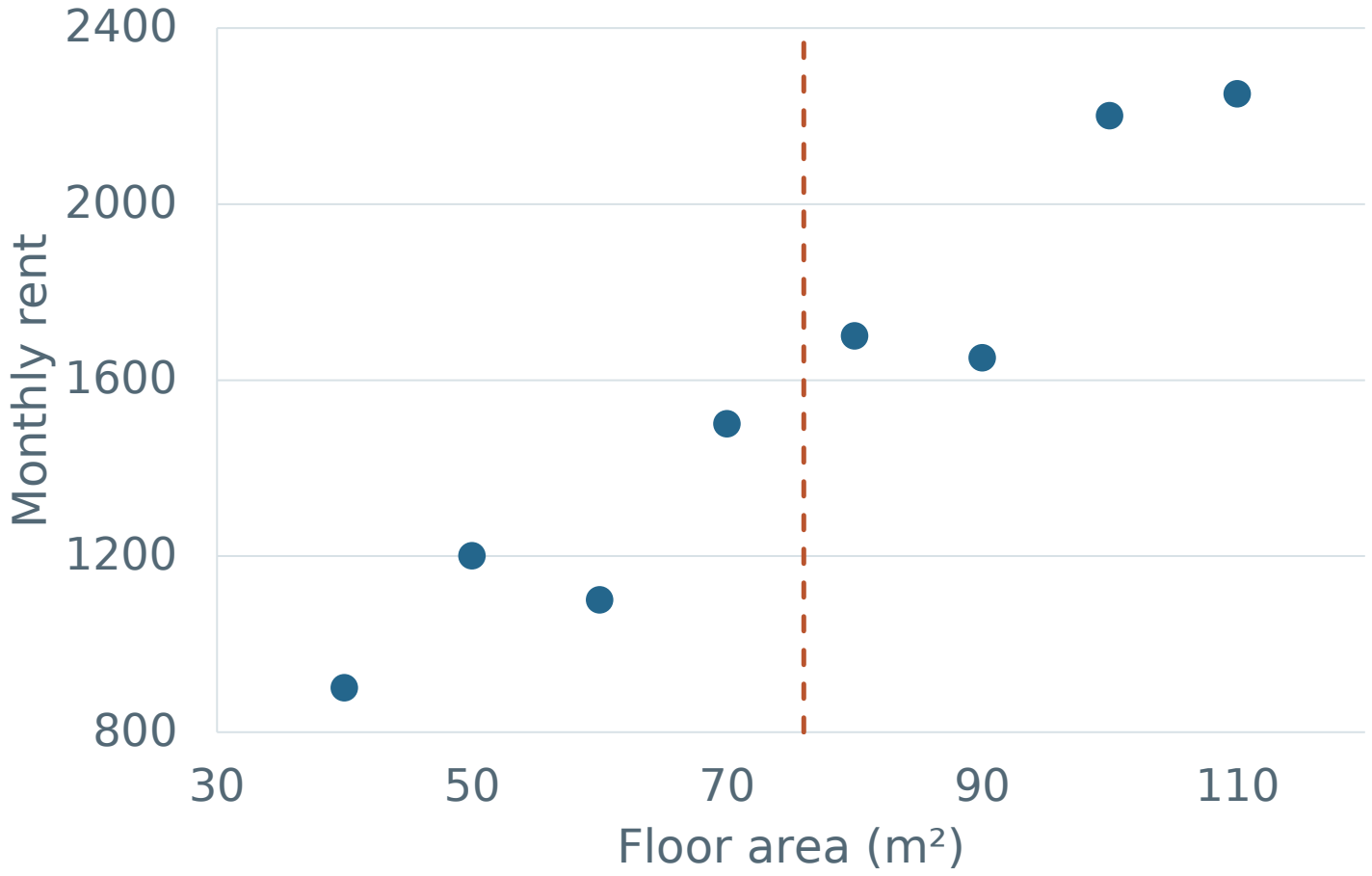
Input x

The features we choose to record

Target y

The quantity we want to predict

A prediction for a new apartment



76 m²

What rent would you predict?

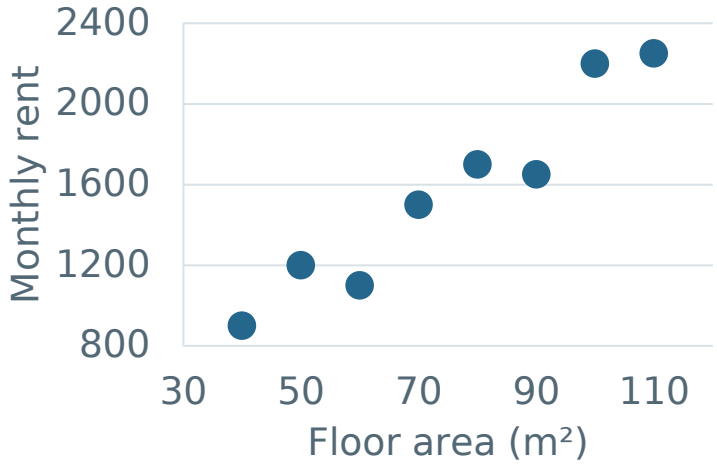
What connects this apartment
to the observed cases?

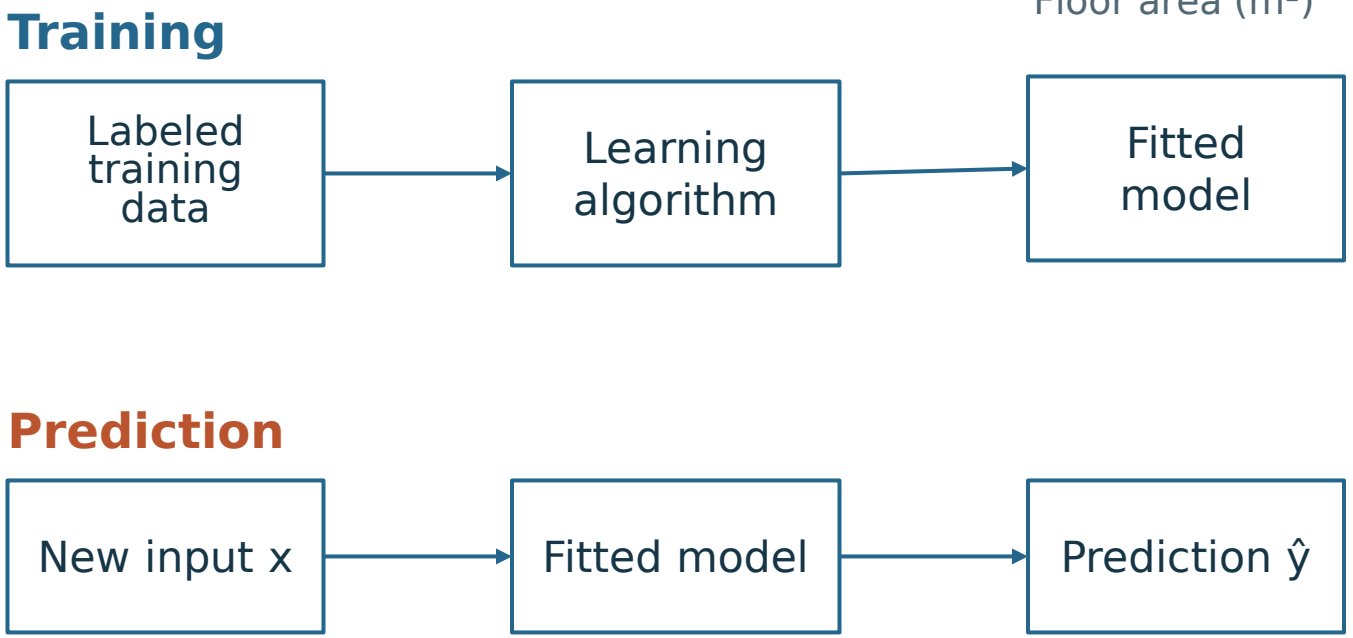
A model turns past examples into a rule for a new case.

Training data contains inputs and labels

Area (m²)	Monthly rent
40	900
50	1200
60	1100
70	1500
80	1700
90	1650
100	2200
110	2250

Label y: the known answer (monthly rent)
Labeled data: examples containing both input x and label y





Memorization handles exact matches

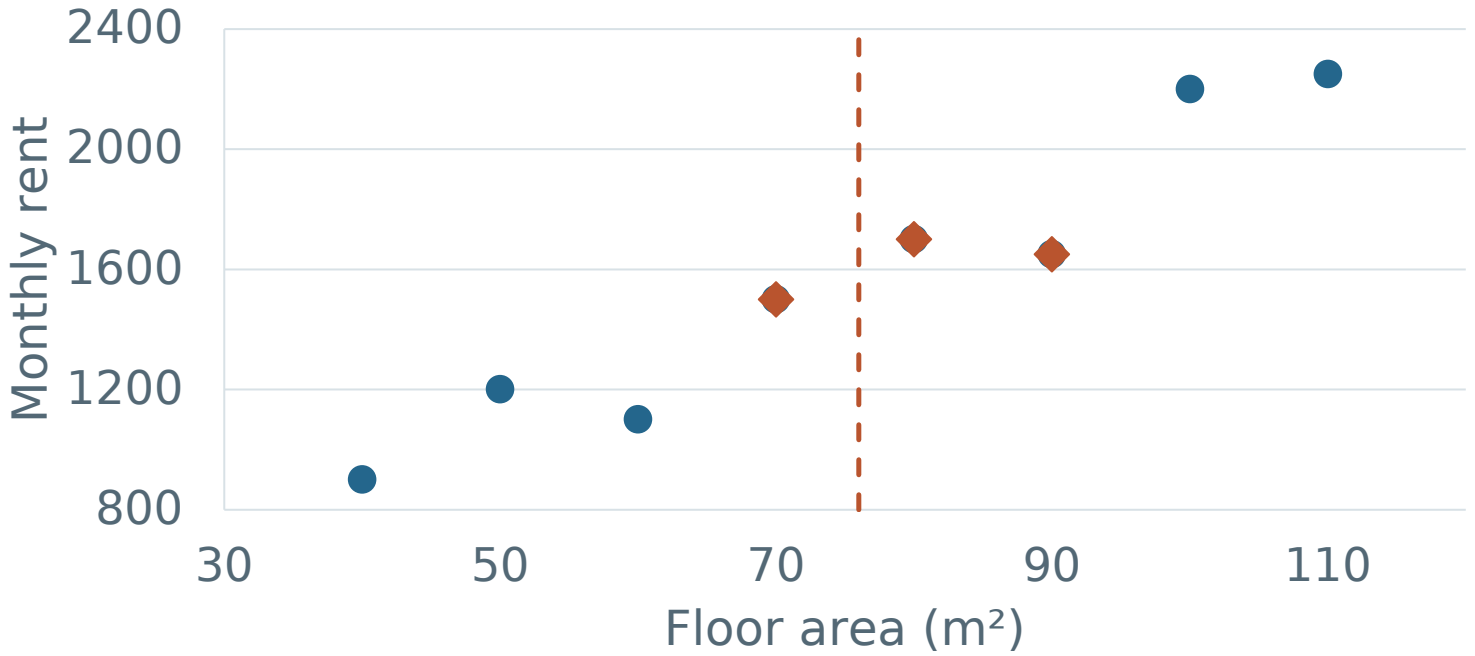
Query area	Lookup result
70 m²	1500
80 m²	1700
79 m²	**No stored answer**

Every training input has an answer

A new input needs a prediction rule

Generalization needs a modeling assumption

Apartments with similar floor areas may have similar rents.



Generalization

Useful predictions for relevant new cases

K-nearest neighbors: from one to several

Distance: $d(x, x_i) = |x - x_i|$

x = 76 m²

k = 1

Use the single nearest training example

Training area	Distance	Observed rent
70	6	1500
80	**4**	**1700**
90	14	1650

Nearest area = 80 m²

ŷ = 1700

k = 3

Average the three nearest examples

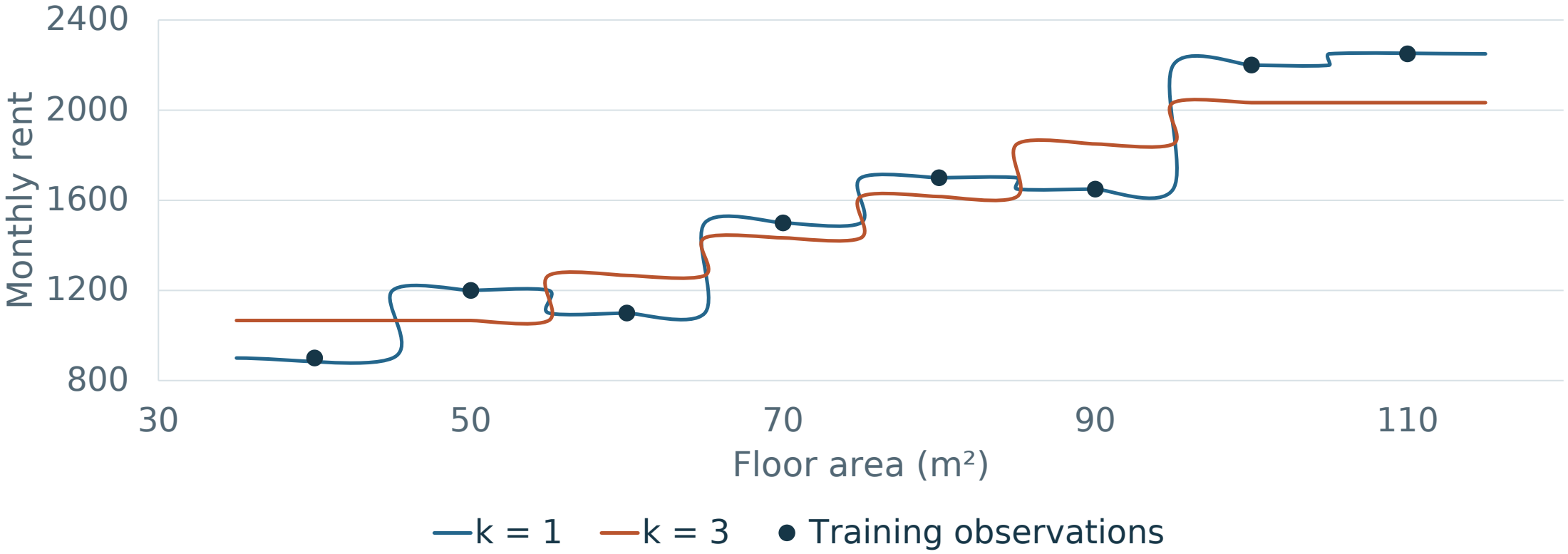
Neighbor order	Area (m²)	Observed rent
1	80	1700
2	70	1500
3	90	1650

ŷ = (1700 + 1500 + 1650) / 3

ŷ ≈ 1616.7

The fitted rule depends on k

$$\widehat{f_k}(x) = \left(\frac{1}{k}\right)\sum y_i \ \text{, for } i \in N_k(x)$$



Which one is better:
k=1 or k=3

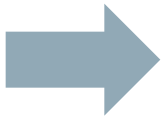
Evaluation estimates performance on future data

Training scores are optimistic: the model has seen that data

THE QUESTION

Will it work on new apartments?

Future data are not available yet.



1 **UNSEEN, LABELED DATA**

New inputs x • Known labels y • Future-like cases

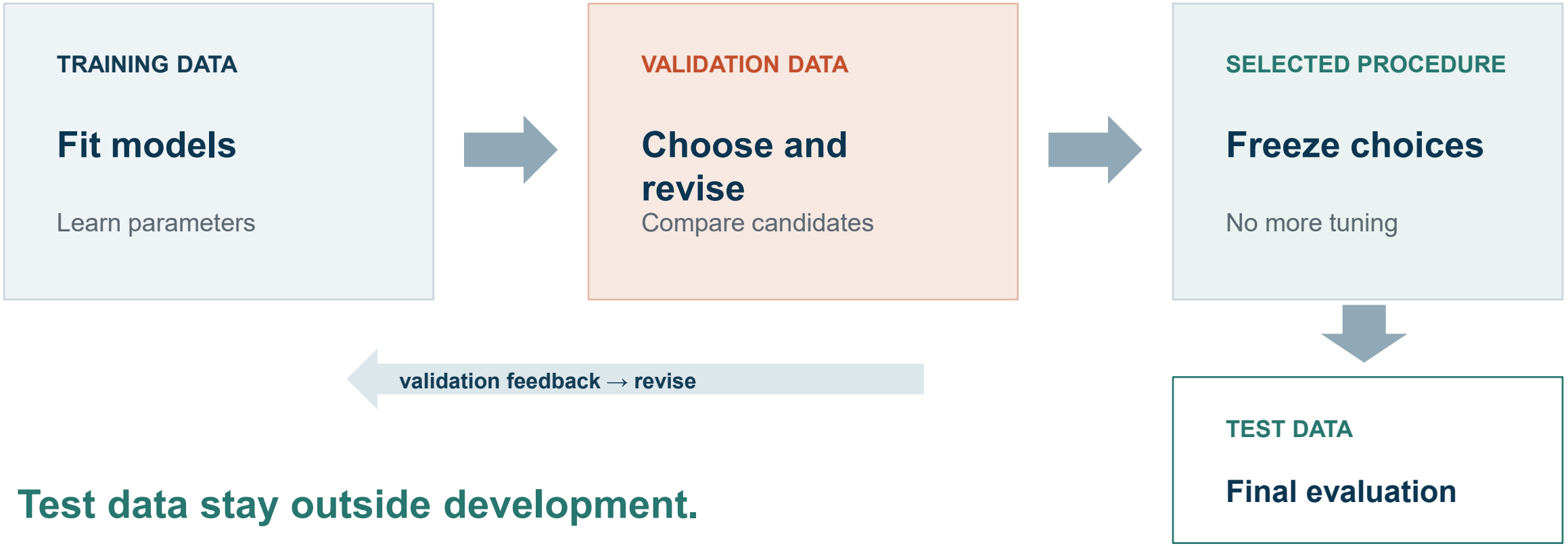
2 **A SUITABLE EVALUATION MEASURE**

One number to score predictions and compare models

Fair evaluation = unseen labeled data + the right measure

Training, validation, and test data have different roles

Validation guides development. Test measures the final result.



Test data stay outside development.

Different tasks use different evaluation measures

The metric must match the prediction task

REGRESSION

Predict a numeric value

Rent, temperature, demand

MAE	average absolute error
MSE	average squared error
RMSE	square root of MSE

CLASSIFICATION

Predict a category

Spam, fraud, image class

Accuracy	fraction correct
Precision / recall	different error types
Log loss	probability quality

Choose a metric that matches the task and the cost of mistakes.

MSE is one evaluation metric for regression

Average squared error across unseen labeled cases

For each evaluation case

squared error $(y - \hat{y})^2$

Exact prediction --> error = 0

Across n evaluation cases

MSE = 1/n × sum of all squared errors

Mean = add the squared errors, then divide by n

What MSE emphasizes

Large errors matter more

Squaring amplifies large misses.

Lower is better

Compare models on the same data.

Not the only regression metric

MAE weights misses linearly.

Validation data guide model selection

Candidate models; choose with validation scores

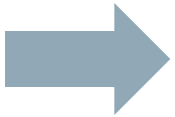
TRAINING DATA

Fit candidate models

k = 1
candidate A

k = 3
candidate B

Both models use the training data.



VALIDATION DATA

Score both on the same validation data

k = 1	MSE_1
k = 3	MSE_3

lower wins

Model selection Choose k from validation—not training performance.

Choose k using validation data

Use cases that are separate from the 76 m² prediction example

Area (m²)	Observed rent	k = 1 prediction	Squared error	k = 3 prediction	Squared error
54	1100	1200	10,000	1066.7	1111.1
78	1600	1700	10,000	1616.7	277.8
96	1950	2200	62,500	2033.3	6944.4

k = 1

(10,000 + 10,000 + 62,500) / 3

MSE = 27,500

k = 3

(1111.1 + 277.8 + 6944.4) / 3

MSE ≈ 2777.8

SELECTED ON THIS VALIDATION SET

A separate test data

Area (m²)	Observed rent	Selected k = 3 prediction
64	1300	1266.7
84	1650	1616.7
104	2100	2033.3

Test MSE ≈ 2222.2

A finite-sample estimate of performance on new cases

Classification using K-NN

Same eight apartments. New target: rented within one week (Quick) or no (Later).

Area (m²)	Class target
40	Later
50	Later
60	**Quick**
70	**Quick**

Area (m²)	Class target
80	**Quick**
90	Later
100	Later
110	Later

Use floor area to predict a class label instead of a rent amount.

The class targets are observed rental outcomes.

Classification predicts by neighbor vote

Query apartment: 76 m² — compare k = 1 and k = 3

Order	Area	Distance	Class	Used by
1	**80**	**4**	**Quick**	**k=1, k=3**
2	70	6	Quick	k=3
3	90	14	Later	k=3

k = 1

nearest class

Quick

k = 3

vote: 2 Quick, 1 Later

Quick

Same prediction here k = 1 uses one sample; k = 3 uses the three nearest samples.

Validation selects k; test evaluates the final model

Both sets are labeled and unseen during training—but they have different roles.

VALIDATION SET

Compare k = 1 and k = 3

Area	Actual	k = 1	k = 3
54	Later	Quick ✕	Later ✓
78	Quick	Quick ✓	Quick ✓
96	Later	Later ✓	Later ✓

k = 1: 2/3 **k = 3: 3/3 → select**

TEST SET

Evaluate only the selected k = 3

Area	Actual	k = 3
60	Later	Later ✓
86	Quick	Quick ✓
100	Later	Later ✓

Final test result: 3/3 correct

Validation informs the choice. Test data are used once after the choice is fixed.

A single data split creates a tradeoff—and some luck

We need enough data to learn, enough to choose, and enough for a trustworthy final test.

ONE COMMON SPLIT



The tradeoff

More training data	better fitted models
More validation data	more reliable model choice
More test data	more reliable final estimate

THE COMPLICATION

One held-out evaluation split may favor one model

Split A	**k = 1: 3/3**
	k = 3: 2/3
Split B	k = 1: 2/3
	k = 3: 3/3

Different evaluation cases → different winner

k-fold cross-validation rotates the evaluation role

Hold out a different fold each round, evaluate each candidate, then average the results.

5 FOLDS OF THE DEVELOPMENT DATA



Each row is one training/evaluation round

AVERAGE EVALUATION SCORE

k = 1

Fold scores

67, 100, 67, 67, 100%

average = 80%

k = 3

Fold scores

100, 67, 100, 100, 100%

average = 93% → select

Cross-validation can support model selection or estimate test performance; its role depends on the evaluation design.

Binary metrics and confusion matrix

Generic example: 200 cases, where “positive” and “negative” classes

Actual \ Predicted	Positive	Negative
Positive	TP = 40	FN = 20
Negative	FP = 10	TN = 130

ACCURACY

(TP + TN) / all cases

170 / 200 = 85%

ERROR RATE

(FP + FN) / all cases

30 / 200 = 15%

Caution **Accuracy can look high when one class is much more common than the other.**

Classification metrics answer different questions

Use the same confusion matrix: TP 40, FP 10, FN 20, TN 130.

Metric	Question	Formula	Value
Precision	When we predict positive, how often are we right?	$TP \,/\, (TP + FP)$	80.0%
Recall / sensitivity	Of all positives, how many did we find?	$TP \,/\, (TP + FN)$	66.7%
F1 score	How strong is the precision–recall balance?	$2PR \,/\, (P + R)$	72.7%
MCC	+1 perfect, 0 random, −1 opposite	$\frac{(TP \cdot TN - FP \cdot FN)}{\sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}}$	0.63

Which one is sensitive to swapping P/N class labels?

Self-training tasks

1	**Distance and scaling**	Study how feature scaling affects neighbors
2	**K-NN design choices**	Study k, how changing k in its range affects the general behavior of the model.
3	**Evaluation pitfalls**	Compare MCC and F1 score. How they behave when we swap P/N class labels.
4	**House-rent project**	Use house_rent_dataset from course page to implement a knn family of models. Select $1 \leq k < 30$ with 5-fold CV; Evaluate the selected model using 5-fold CV.